Table 15.6 Tissue weighing factors, w_T (ICRP 2007)

Body part	w_T	$\Sigma\, w_T$
Bone-marrow (red), colon, lung, stomach, breast, remainder	0.12	0.72
Gonads	0.08	0.08
Bladder, oesophagus, liver, thyroid	0.04	0.16
Bone surface, brain, salivary glands, skin	0.01	0.04
Total		1.00

$\Sigma\, w_T$ = sum of w_T-values. Remainder = Adrenals, extrathoracic region, gall, bladder, heart, kidneys, lymphatic nodes, muscle, oral mucosa, pancreas, prostate, small intestine, spleen, thymus, uterus/cervix

sizes. Table 15.5 gives doses which are inactivating or lethal to organisms within 30 days ($LD_{50/30}$) after short-time irradiation. A 10-fold dose is required to kill rather than to inactivate microorganism cells.

Because of the varying radiation sensitivities of the different cell types, a *tissue weighing factor*, w_T, is introduced. It represents the relative contribution of that tissue to the total detriment from uniform irradiation of the body; Table 15.6 gives tissue weighing factors. This leads to another dose concept (cf. eqn. 15.1), the *effective equivalent dose*, H_E, which is the sum of the weighted equivalent doses in all tissues, as given by

$$H_E\,(\mathrm{Sv}) = \sum_T w_T H_T \tag{15.2}$$

The weighing factors are important in medical radiation therapy and e.g. for evaluating the effects of internal radionuclides, §15.13.5. For low-LET whole-body irradiation w_R and $\Sigma w_T = 1$, i.e. the indici in (15.1) and (15.2) may be dropped.

15.4. SOME CONCEPTS OF RADIATION BIOLOGY

In radiology, the biological effects of radiation are usually discussed along two lines: the matrix effect and target theory. The *matrix effect* considers the particle-water interaction in which ions, radicals, and excited atoms are produced. This is the dominating effect at large radiation doses and dose rates; the lethal dose of 10 Gy generates about 10 000 electron tracks in each human cell. Free radicals and oxidizing products interact directly with cell DNA, causing the DNA-strands to break as described in §15.2.2. One can state that at such high doses the cell is simply poisoned by decomposition products, and the whole organ may be destroyed. It should be noted that in cancer therapy even much higher doses are given.

Inactivation of cell populations (Figure 15.2) and organisms (Table 15.5) requires comparatively large doses and the matrix effect should dominate. However, a 10 Gy dose to cells 1/1000 the size of human cells produces only 10 electron tracks per cell. The cell repair mechanisms are able to overcome the poisonous products. Consequently, doses

required for inactivation of enzymes and viruses must be 10^5-10^6 times larger than the ones that are lethal to man.

Target theory depicts the DNA molecule as the site of reaction. Even if matrix effects can never be excluded, target theory is the essential model at low levels of irradiation because a single change in a DNA molecule, however unlikely, may convert it into an *oncogene* (tumor producing gene) with fatal consequence to the organism. Natural background radiation is mainly low-LET. At a common level of 1 mGy/y, each human cell receives on average one track intersection per year. Nevertheless, it is claimed (e.g. UNSCEAR 1993) that this low-level low-LET radiation may be responsible for many of the malignant cancers in the population, though no laboratory experiments have confirmed this, as larger doses are required to produce observable laboratory results.

Figure 15.3 shows typical *dose-response curves*: the effects of X-ray irradiation of hamster cells (*in vitro*) at different stages of the cell cycle. The surviving fraction of cells decreases with increasing dose. Cells in the late G_2 stage and in mitosis are more sensitive than cells at G_1 and the S stage. This is a general observation, which can be explained by assuming

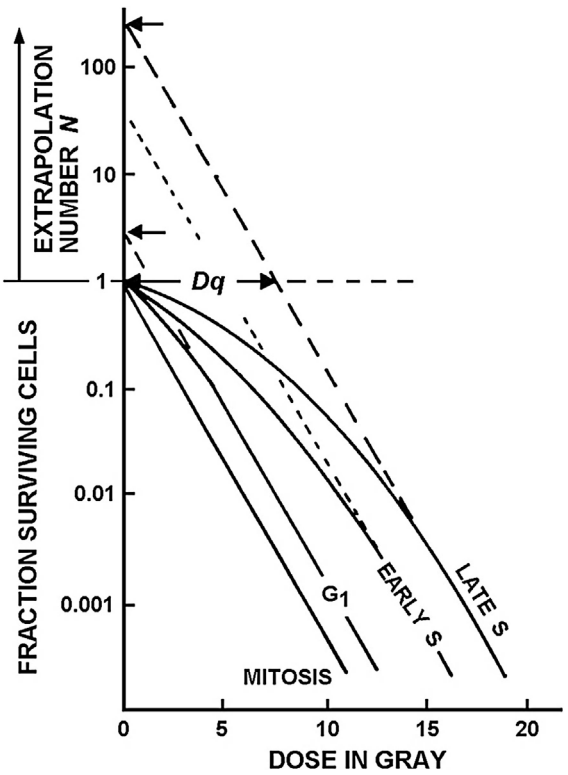


Figure 15.3 Survival curves for Chinese hamster cells at different stages of the cell cycle. *(Sinclair and Morton 1966, from Nias.)*

that the likelihood for double strand breaks of the DNA is greater and the cell repair mechanism less efficient when the DNA is in the condensed chromosome state as compared to when it is in the chromatin state (S-state). In Figure 15.3, the response function for mitosis seems to be linear while the "late S" function is curved. Extrapolation of the straight part of the S-curve extends it to the zero dose line at ~100; this number is referred to as the *extrapolation number*, *N*. Its significance is not quite clear. The typical values for human cells and X- and γ-radiation are 2−10. The value of the extrapolated line at 100% cell survival (or $N = 1$) is referred to as the "*quasi-threshold dose*", D_q. The "shoulder" in the dose-response curve is typical for X- and γ-irradiation of human cells while neutrons and alpha particles hardly produce any shoulder. Also, a shoulder does not form when very simple organisms, such as viruses, are irradiated by X- and γ-rays. A simple straight dose-response relation (the mitosis line) is taken as support of a *single-cell hit killing*, which is purely random. We have already concluded that n and α radiations are likely to severely damage DNA, particulary during the G_2 and mitosis stages.

The existence of a shoulder is taken as support of a *multi-target model*: the DNA must be damaged at several points by X- or γ-rays in order to cause the cell to die because of the efficient repair mechanism for single-strand breaks.

The slope of the straight part of the line allows calculation of the *mean lethal dose*, D_0, which is the dose required to inactivate the fraction $1 - e^{-1}$ (i.e. 63%) of the cells. Designating the surviving fraction as f_S gives

$$f_S = e^{-D/D_0} \tag{15.3}$$

for the straight line (mitosis). The curve can be described by

$$f_S = 1 - \left(1 - e^{D/D_0}\right)^N \tag{15.4}$$

15.5. FURTHER REGULARITIES AT LARGE DOSES

Figure 15.4 shows the number of dicentric chromosome abberations observed in a cell population as a function of neutron or gamma/X-ray energy. *Dicentric abberations* (Fig. 15.1.D) seem to be the most consistent index of radiation damage and represents about 60% of all observable unstable abberations following acute irradiation. Their background frequency is very low (about 1 in 1000 lymphocyte cells). From Figure 15.4, it is seen that the lower the radiation particle energy is, the larger the damage (of course, until the energy has decreased to such low levels that its ionization effects begin to drop abruptly, see Figure 7.7). This is attributed to the increasing ionization density with decreasing particle energy, as described in chapter 7; see also Tables 15.2 and 15.4.

Figure 15.4 also shows that the damage increases with increasing *dose rate*. This is a general and significant phenomenon which is probably related to the cell (DNA) repair